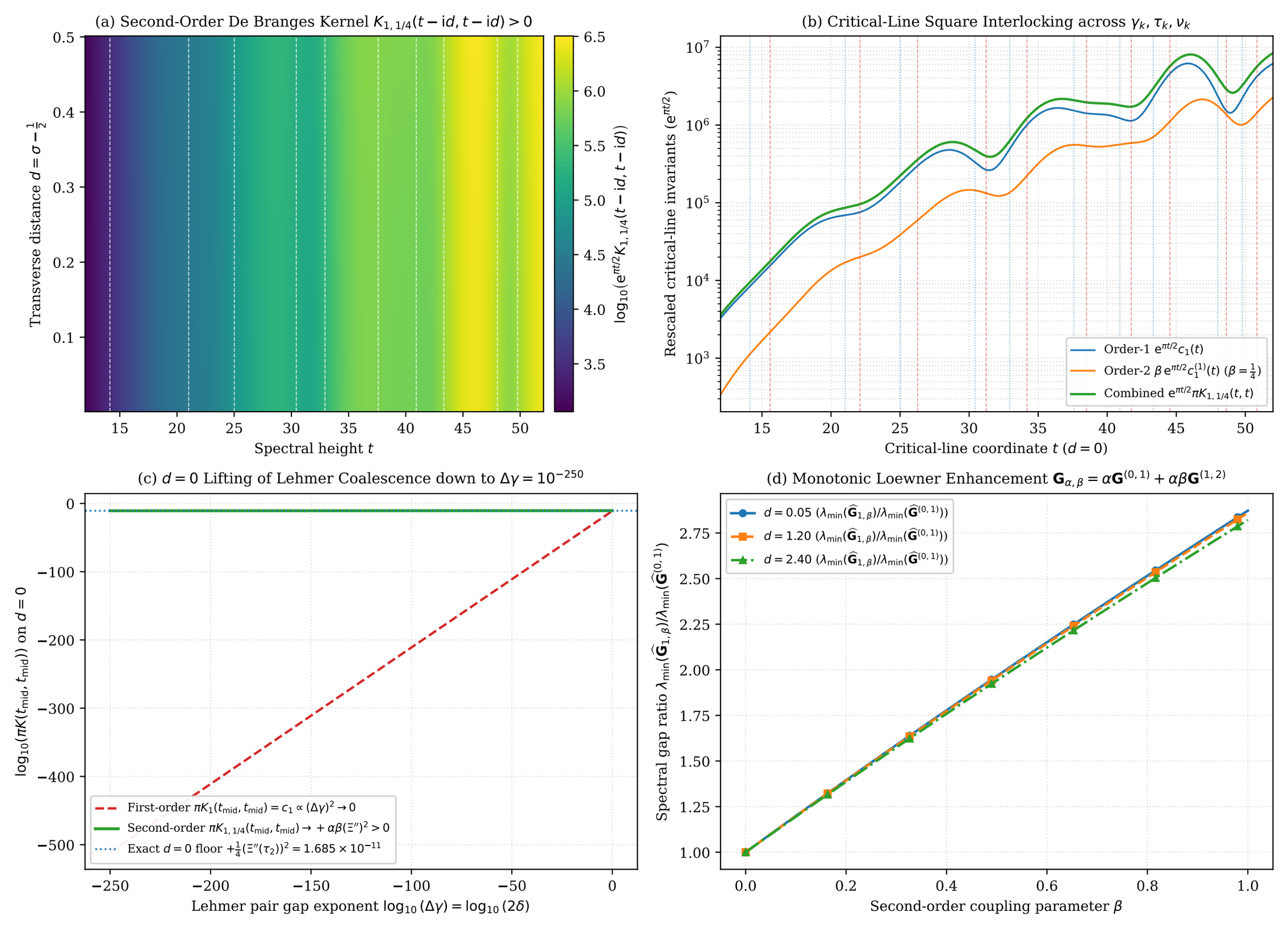

(a) Second-Order De Branges Kernel $K_{1,1/4}(t-id, t-id) > 0$
Transverse distance $d = \sigma - \frac{1}{2}$
Spectral height t
$\log_{10}(e^{\pi t/2} K_{1,1/4}(t-id, t-id))$
(b) Critical-Line Square Interlocking across γ_k, τ_k, ν_k
Rescaled critical-line invariants ($e^{\pi t/2}$)
Critical-line coordinate t ($d = 0$)
Order-1 $e^{\pi t/2} c_1(t)$
Order-2 $\beta e^{\pi t/2} c_1^{(1)}(t)$ ($\beta = \frac{1}{4}$)
Combined $e^{\pi t/2} \pi K_{1,1/4}(t, t)$
(c) $d = 0$ Lifting of Lehmer Coalescence down to $\Delta\gamma = 10^{-250}$
$\log_{10}(\pi K(t_{mid}, t_{mid}))$ on $d = 0$
Lehmer pair gap exponent $\log_{10}(\Delta\gamma) = \log_{10}(2\delta)$
First-order $\pi K_1(t_{mid}, t_{mid}) = c_1 \propto (\Delta\gamma)^2 \to 0$
Second-order $\pi K_{1,1/4}(t_{mid}, t_{mid}) \to +\alpha\beta(\Xi'')^2 > 0$
Exact $d = 0$ floor $+\frac{1}{4}(\Xi''(\tau_2))^2 = 1.685 \times 10^{-11}$
(d) Monotonic Loewner Enhancement $\mathbf{G}_{\alpha,\beta} = \alpha\mathbf{G}^{(0,1)} + \alpha\beta\mathbf{G}^{(1,2)}$
Spectral gap ratio $\lambda_{min}(\widehat{\mathbf{G}}_{1,\beta})/\lambda_{min}(\widehat{\mathbf{G}}^{(0,1)})$
Second-order coupling parameter β
$d = 0.05$ $(\lambda_{min}(\widehat{\mathbf{G}}_{1,\beta})/\lambda_{min}(\widehat{\mathbf{G}}^{(0,1)}))$
$d = 1.20$ $(\lambda_{min}(\widehat{\mathbf{G}}_{1,\beta})/\lambda_{min}(\widehat{\mathbf{G}}^{(0,1)}))$
$d = 2.40$ $(\lambda_{min}(\widehat{\mathbf{G}}_{1,\beta})/\lambda_{min}(\widehat{\mathbf{G}}^{(0,1)}))$